Problem-Solution Fit

Date	02 October 2026
Team ID	SWTID-2026-3981
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Students of all academic levels (beginners, school students, self-learners) Educational institutions and content platforms seeking AI-powered learning tools. Developers looking for lightweight, accessible generative Al setups.	**6. CUSTOMER LIMITATIONS** EG. BUDGET, DEVICES **[CL]** Hardware limitations (expensive GPUs and reliance on standard laptops or Mac M1 devices). High internet bandwidth or heavy cloud cost constraints. Time constraints for reading long passages.	5. **AVAILABLE SOLUTIONS PROS & CONS [AS]** **Pros:** Traditional learning sources & heavy AI tools offer detailed information. **Cons:** Complex language, high computing resource requirements, lack of personalization, and slow processing standard hardware.

2. PROBLEMS / PAINS
+ ITS FREQUENCY **[PR]**

Difficulty understanding complex concepts.

Difficulty summarizing large educational passages, especially when studying a country's history and timeline.

Lack of personalized quizzes and structured learning paths during practice sessions.

Frequency: A recurring issue faced during everyday study sessions.

9. PROBLEM ROOT / CAUSE [RC]

Traditional study content can be complex, while access to personal tutoring is limited.

Existing AI tools require heavy hardware, making them difficult for everyone to access directly.

7. BEHAVIOR
+ ITS INTENSITY **[BE]**

Students prefer to learn concepts through easily digestible summaries and step-by-step guidance.

High intensity: Students expect instant doubt-solving and active practice through quizzes.

3. TRIGGERS TO ACT [TR]

Exams or tests create a need for instant clarification and support.

Long textbook paragraphs are overwhelming and confusing.

A specific topic (e.g., SQL or the Pythagorean Theorem) is easier to learn through a guided learning path.

10. YOUR SOLUTION [SL]

EduGenie: A lightweight AI-powered educational assistant driven by generative AI.

Google Gemini 1.5 Pro API and LaMini-Flan-T5 models combine instant Q&A, concept simplification, quiz generation, text summarization, and personalized learning paths to provide a seamless learning experience.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

Web interface (FastAPI + HTML/CSS), RESTful API end points (/qa, /explain, /quiz, /summarize, /learn/recommendations).

4. EMOTIONS
BEFORE / AFTER **[EM]**

Before: Overwhelmed, confused, and frustrated with static and heavy study materials.

After: Confident, clear, and motivated with interactive, simplified, and instant AI guidance.

OFFLINE

Local CPU/hardware inference using lightweight models like LaMini-Flan-T5.